

Econometrics

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Week 5 Notes

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1 Panel Data

1.1 Introduction

A panel or longitudinal data set is one where there are repeated observations on the same units, individuals, household firms, countries, or any set of entities that remain stable over time.

1.1.1 Balanced and Unbalanced data sets

Notation:

$$Y_{i,t}, i = 1, \dots, N; t = 1, \dots, T$$

A panel is balanced if all units are observed in all periods the notation is

$$NT$$

If some units are missing in some periods the panel is unbalanced:

$$\sum_{i=1}^N T$$

With panel data we can study different issues:

- *cross sectional variation* unobservable in time series data
- *time series variation* unobservable in cross sectional data
- Heterogeneity
- Hierarchical structures say city state etc
- Dynamics in economic behavior
- Individual/group effects
- Time effects

1.2 Panel Data Models

Panel data models describe the individual behavior both across time and across individuals.

There are three types of models:

- The pooled model
- Fixed effects model
- Random effects model

1.2.1 Pooled model

The pooled model specifies constant coefficients, the usual assumptions for cross sectional analysis

$$Y_{it} = \alpha + X'_{it}\beta + u_{it}$$

This is the most restrictive panel data model and is not used much in the literature.

The pooled model is a pooled linear regression without fixed and random effects.

It assumes a constant intercept and slope regardless of group and time period.

```
library(plm)
data("Grunfeld")
pooling <- plm(inv ~ value + capital, data = Grunfeld, model = "pooling", index = c("firm", "year"))
broom::tidy(pooling)
```

```
## # A tibble: 3 x 5
##   term      estimate std.error statistic  p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept) -42.7      9.51     -4.49 1.21e- 5
## 2 value        0.116    0.00584    19.8 9.54e-49
## 3 capital      0.231    0.0255     9.05 1.35e-16
```

We check if the data is balanced as follows:

```
library(plm)
Grunfeld %>%
  is.pbalanced()
```

```
## [1] TRUE
```

1.2.2 Individual-specific effects Model

We assume that there is unobserved heterogeneity across the individuals captured by α_i

The main question is whether the individual specific effects α_i are correlated with the regressors.

If they are correlated we have fixed effects model. If they are not correlated we have random effects model.

1.2.3 Fixed Effects Model

The FE model allows the individual-specific effects α_i to be correlated with the regressor x .

We include α_i as intercept.

Each individual has a different intercept term and the same slope parameters

$$Y_{it} = \alpha_i + X'_{it}\beta + u_{it}$$

We can recover the individual specific effects after estimation as:

$$\hat{\alpha}_i = \bar{y}_i - \bar{x}_i \hat{\beta}$$

Variables that change little or not at all over time such as individual characteristics should not be included in a fixed effects model because they produce collinearity with fixed effects.

```
library(plm)
model_fixed <- plm(inv ~value + capital, data = Grunfeld, model = "within", index = c("firm","year"))
broom::tidy(model_fixed)
```

```
## # A tibble: 2 x 5
##   term      estimate std.error statistic  p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 value      0.110     0.0119      9.29 3.92e-17
## 2 capital    0.310     0.0174     17.9 2.22e-42
```

Testing if the fixed effects are necessary we compare the fixed effects model to the pooled model.

We test the hypothesis

Null Hypothesis: No Fixed Effects

Alternative: There are fixed effects

```
broom::tidy(pFtest(model_fixed, pooling))
```

```
## Multiple parameters; naming those columns df1, df2
## # A tibble: 1 x 6
##   df1  df2 statistic  p.value method alternative
##   <int> <int>    <dbl>    <dbl> <chr>      <chr>
## 1     9   188    49.2 8.70e-45 F test for individual effects significant effe~
```

The p-value is less than 0.05 and thus the null hypothesis is rejected.

1.2.4 Random Effects Model

The RE model assumes that the individual specific effects α_i are distributed independently of the regressors.

We include α_i in the error term.

Each individual has the same slope parameter and a composite error term

$$\epsilon_{it} = \alpha_{it} + e_{it}$$

$$Y_{it} = X_{it}\beta + (\alpha_i + e_{it})$$

Here

$$Var(\epsilon_{it}) = \sigma_{\alpha}^2 + \sigma_e^2 \text{ and } Cov(\epsilon_{it}, \epsilon_{is}) = \sigma_{\alpha}^2$$

```
library(plm)
model_random <- plm(inv ~value + capital, data = Grunfeld, model = "random", index = c("firm","year"))
broom::tidy(model_random)
```

```
## # A tibble: 3 x 5
##   term      estimate std.error statistic  p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
```

```
## 1 (Intercept)  -57.8      28.9      -2.00 4.54e- 2
## 2 value        0.110     0.0105     10.5  1.28e-25
## 3 capital      0.308     0.0172     17.9  6.41e-72
```

1.2.5 Choosing between fixed and random effects

1.2.5.1 Breusch Pagan Lagrange Multiplier test

- This is a test for the random effects model based on the OLS residual.
- Test whether σ_u^2 is significantly different from zero
- If the LM test is significant use the random effects instead of the OLS model

```
broom::tidy(pFtest(model_fixed, model_random))
```

```
## Multiple parameters; naming those columns df1, df2
```

```
## # A tibble: 1 x 6
```

```
##   df1  df2 statistic p.value method alternative
##   <int> <int>      <dbl>  <dbl> <chr>      <chr>
## 1     9   188      1.01   0.430 F test for individual effects significant effec~
```

1.2.5.2 Hausman Test

- The random effects estimator is more efficient so we need to use it if the Hausman test supports it. If it does not support it, use the fixed effects model.
- Hausman test tests whether there is a significant difference between the fixed and random effects estimators.
- The null hypothesis is that random effects model is better while alternative is that the fixed effects model is better.

```
phptest(model_random, model_fixed)
```

```
##
```

```
## Hausman Test
```

```
##
```

```
## data: inv ~ value + capital
```

```
## chisq = 2.3304, df = 2, p-value = 0.3119
```

```
## alternative hypothesis: one model is inconsistent
```

The fixed effects estimator will always give consistent estimates, but they may not be the most efficient.

The random effects estimator is inconsistent if the appropriate model is the fixed effects model.

The random effects estimator is consistent and most efficient if the appropriate model is random effects model.